

Optical simulation of photovoltaic modules with multiple textured interfaces using the matrix-based formalism OPTOS

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Abstract: The OPTOS formalism is a matrix-based approach to determine the optical properties of textured optical sheets. It is extended within this work to enable the modelling of systems with an arbitrary number of textured, plane-parallel interfaces. A matrix-based system description is derived that accounts for the optical reflection and transmission interaction between all textured interfaces. Using OPTOS, we calculate reflectance and absorptance of complete photovoltaic module stacks, which consist of encapsulated silicon solar cells featuring textures that operate in different optical regimes. As exemplary systems, solar cells with and without module encapsulation are shown to exhibit a considerable absorptance gain if the random pyramid front side texture is combined with a diffractive rear side grating. A variation of the sunlight's angle of incidence reveals that the grating gain is almost not affected for incoming polar angles up to 60°. Considering as well the good agreement with alternative simulation techniques, OPTOS is demonstrated to be a versatile and efficient method for the optical analysis of photovoltaic modules.

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OCIS codes: (040.5350) Photovoltaic; (050.1950) Diffraction gratings; (080.1753) Computation methods.

References and links

1. R. Brendel, "Sunrays. A versatile ray tracing program for the photovoltaic community," in *Proceedings of the 12th European Photovoltaic Solar Energy Conference* (1994), pp. 1339–1342.
2. S. C. Baker-Finch and K. R. McIntosh, "Reflection of normally incident light from silicon solar cells with pyramidal texture," *Prog. Photovolt. Res. Appl.* **19**(4), 406–416 (2011).
3. H. Holst, M. Winter, M. R. Vogt, K. Bothe, M. Köntges, R. Brendel, and P. P. Altermatt, "Application of a new ray tracing framework to the analysis of extended regions in Si solar cell modules," *Energy Procedia* **38**, 86–93 (2013).
4. J. M. Greulich, *Simulation and Characterization of Novel Large-Area Silicon Solar Cells for Industrial Production* (Fraunhofer Verlag, 2014).
5. B. Lipovšek, J. Krč, and M. Topič, "Optical model for thin-film photovoltaic devices with large surface textures at the front side," *Informacije MIDEM* **41**, 1–8 (2011).
6. Z. C. Holman, M. Filipič, B. Lipovšek, S. de Wolf, F. Smole, M. Topič, and C. Ballif, "Parasitic absorption in the rear reflector of a silicon solar cell. Simulation and measurement of the sub-bandgap reflectance for common dielectric/metal reflectors," *Sol. Energy Mater. Sol. Cells* **120**, 426–430 (2014).
7. R. Rothmund, T. Umdum, G. Meinhardt, K. Hingerl, T. Fromherz, and W. Jantsch, "Light trapping in pyramidally textured crystalline silicon solar cells using back-side diffractive gratings," *Prog. Photovolt. Res. Appl.* **21**, 747–753 (2012).
8. R. Santbergen and R. van Zolingen, "The absorption factor of crystalline silicon PV cells. A numerical and experimental study," *Sol. Energy Mater. Sol. Cells* **92**(4), 432–444 (2008).
9. R. Santbergen, A. H. M. Smets, and M. Zeman, "Optical model for multilayer structures with coherent, partly coherent and incoherent layers," *Opt. Express* **21**(S2 Suppl 2), A262–A267 (2013).

10. A. Mellor, H. Hauser, C. Wellens, J. Benick, J. Eisenlohr, M. Peters, A. Guttowski, I. Tobias, A. Marti, A. Luque, and B. Bläsi, "Nanoimprinted diffraction gratings for crystalline silicon solar cells: implementation, characterization and simulation," *Opt. Express* **21**(S2 Suppl 2), A295–A304 (2013).
11. A. Mellor, I. Tobias, A. Marti, M. J. Mendes, and A. Luque, "Upper limits to absorption enhancement in thick solar cells using diffraction gratings," *Prog. Photovolt. Res. Appl.* **19**(6), 676–687 (2011).
12. J. Eisenlohr, N. Tucher, O. Höhn, H. Hauser, M. Peters, P. Kiefel, J. C. Goldschmidt, and B. Bläsi, "Matrix formalism for light propagation and absorption in thick textured optical sheets," *Opt. Express* **23**(11), A502–A518 (2015).
13. N. Tucher, J. Eisenlohr, P. Kiefel, O. Höhn, H. Hauser, M. Peters, C. Müller, J. C. Goldschmidt, and B. Bläsi, "3D optical simulation formalism OPTOS for textured silicon solar cells," *Opt. Express* **23**(24), A1720–A1734 (2015).
14. Y. Li, Y. Chen, Z. Ouyang, and A. Lennon, "Angular matrix framework for light trapping analysis of solar cells," *Opt. Express* **23**(24), A1707–A1719 (2015).
15. M. A. Green, "Self-consistent optical parameters of intrinsic silicon at 300 K including temperature coefficients," *Sol. Energy Mater. Sol. Cells* **92**(11), 1305–1310 (2008).
16. R. Branke and A. Heimsath, "Raytrace3D power tower - A novel optical model for central receiver systems," in *SolarPACES 2010, 16th Solar Power And Chemical Energy Systems International Symposium* (2010).
17. P. Schöttl, P. Kiefel, A. Heimsath, G. Bern, and D. W. van Rooyen, "Novel ray tracing tool for solar applications," in preparation.
18. J. M. Rodriguez, I. Tobias, and A. Luque, "Random pyramidal texture modelling," *Sol. Energy Mater. Sol. Cells* **45**(3), 241–253 (1997).
19. P. Campbell and M. A. Green, "Light trapping properties of pyramidally textured surfaces," *J. Appl. Phys.* **62**(1), 243 (1987).
20. "PV-Lighthouse," www.pvlighthouse.com.au.
21. S. Duttagupta, F. Ma, B. Hoex, T. Mueller, and A. G. Aberle, "Optimised antireflection coatings using silicon nitride on textured silicon surfaces based on measurements and multidimensional modelling," *Energy Procedia* **15**, 78–83 (2012).

1. Introduction

Photovoltaic modules typically consist of interconnected silicon solar cells, which are encapsulated in an ethylene vinyl acetate (EVA) layer and covered by a protective glass at the front side. This module encapsulation can strongly influence the optical properties of the cell's surface texture. The additional interfaces reduce the front side transmission for light incident from the sun but can as well increase the internal reflection towards the solar cell. The refractive index of the medium encapsulating the textured silicon front side can change the transmission angles as well as reflection and transmission efficiencies according to Snell's law and the Fresnel coefficients, respectively.

Optical simulation is an invaluable tool in order to understand the light paths within the photovoltaic (PV) module as well as for the prediction of its optical performance. Ray tracing is the standard modelling approach for textured solar cells and modules, as it is an instructive technique with good agreement between simulation and measurement for textures such as random pyramids [1–4]. One drawback of the classical ray tracing method is its restriction to textures with characteristic size dimensions much larger than the wavelength under consideration. It excludes e.g. diffractive gratings as their optical properties are based on periods in the range of the wavelength. This can be overcome by more advanced approaches, which combine ray tracing with wave optical methods as the transfer matrix method [5,6] or reflectance distribution functions [7]. A remaining disadvantage is the fact that changed parameters usually require a new calculation of the complete system. This is especially time consuming for parameter variations and the investigation of more complex systems, such as photovoltaic module stacks.

Matrix-based methods as published by Santbergen et al. [8,9], Mellor et al. [10,11] or the OPTOS (Optical Properties of Textured Optical Sheets) formalism [12,13] can considerably improve both of these aspects. OPTOS represents the light surface interaction and the propagation through a medium via multiplication of angular redistribution matrices. Once these matrices are calculated, optical properties like reflectance, transmittance and absorptance of a sheet system with two plane-parallel, textured interfaces can be determined with low computational resources. This holds true for the variation of parameters as the sheet

thickness or the angle of incidence, as described in detail for sheets with two textured interfaces like silicon solar cells [12,13].

This work covers the extension of the OPTOS formalism to three or more textured interfaces. It thereby allows for the modelling of photovoltaic module stacks with additional glass or EVA encapsulation layers that could incorporate surface structures as well. Note that by photovoltaic module stack we refer to the layer structure of a module but do not consider area related geometrical properties like spacing between adjacent solar cells. The mathematical description which is partly described in [14] will be introduced in section 2, followed by a detailed investigation of the redistribution matrices in section 3. The last section includes the comparison with an alternative simulation technique as validation of the procedure. The evaluation of a solar cell featuring a random pyramid front and a diffractive grating rear surface at cell and at module level as well as an efficient angle of incidence variation serve as examples of the versatility of the OPTOS formalism in the context of an application relevant case.

2. Methods

2.1 OPTOS module simulation sequence

The system under consideration incorporates two optically thick layers with three textured interfaces, surrounded by air. It is sketched in Fig. 1. In case of a photovoltaic module stack simulation, the bottom layer can be identified with the silicon solar cell bulk and the top layer with glass or EVA.

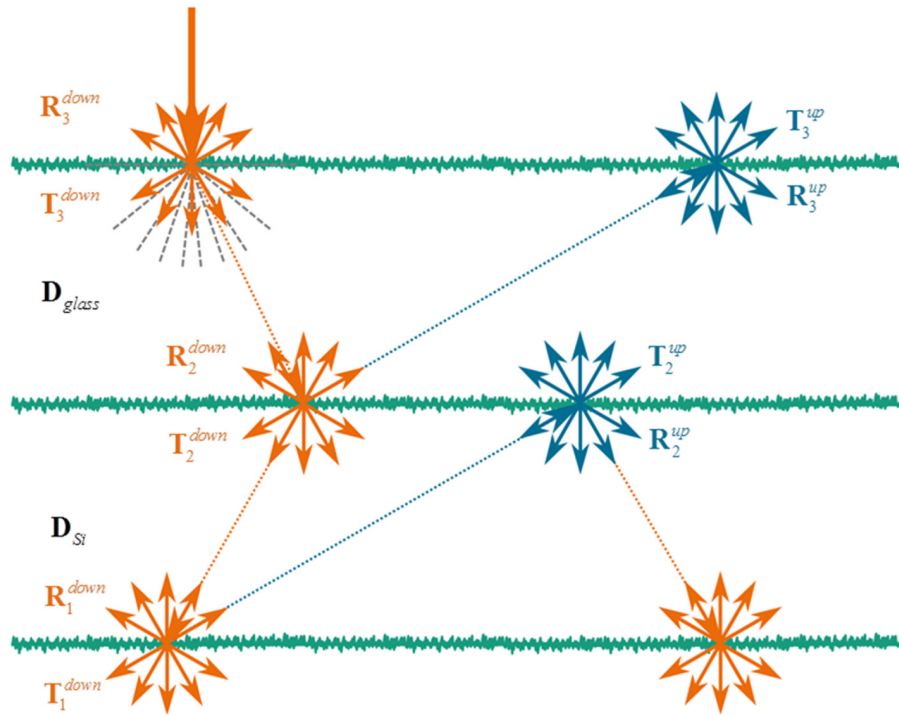


Fig. 1. Schematic view of a system with three textured interfaces. Light impinging from the top is redistributed at each interface into different transmission and reflection angles. The predefined set of angle channels enables the description of these interactions by redistribution matrices of finite size. Including the incoming light directions, “up” (blue) and “down” (orange), there are four distinct matrices which describe each interface.

As in [12,13], the angular half space is discretized into a finite number of angle channels. Light impinging onto a system from a certain angle can be described by a power vector $\mathbf{v}_{inc}$, which has one entry for each angle channel and polarization state. The angular redistribution of the light at each interface is described by matrices. They contain all redistribution information of the transmitted and reflected power with respect to the predefined set of angle channels. Reflection and transmission matrices are named $\mathbf{R}$ and $\mathbf{T}$, respectively. The interface numbering starts at the bottom and is added as subscript to the corresponding matrix name. Since light can impinge on all surfaces from both sides, the incident light direction is denoted with “up” and “down” in the superscript of the matrix name. Altogether there are four distinct matrices to fully describe a single interface interaction. According to the feature sizes of the textures, the matrices can be calculated either with a ray optics or a wave optics approach. The propagation matrices $\mathbf{D}_{glass}$ and $\mathbf{D}_{si}$ are diagonal for both layers but vary according to their absorption coefficients. For more details concerning matrix structure and angular discretization, see [12,13].

2.2 Effective redistribution matrices

The extension of the OPTOS formalism is based on the idea of replacing one layer featuring two textured interfaces with a single effective interface. The resulting effective redistribution matrices can subsequently be used in the formalism as described in [12,13] to calculate the absorptance of one textured layer, e.g. the bottom one. By repeating this procedure, the OPTOS formalism can, in principle, handle arbitrary numbers of interfaces or layers.

The calculation of the effective matrices for an effective interface is done based on the light redistribution of two interfaces and propagation through the enclosed single layer. In general, any layer can be chosen for this calculation. Without loss of generality, we discuss the example depicted in Fig. 2 and start with the effective reflection of the upper layer for light incident from above, $\mathbf{R}_{2,eff}^{down}$ (corresponding to glass/EVA layer in the PV module case).

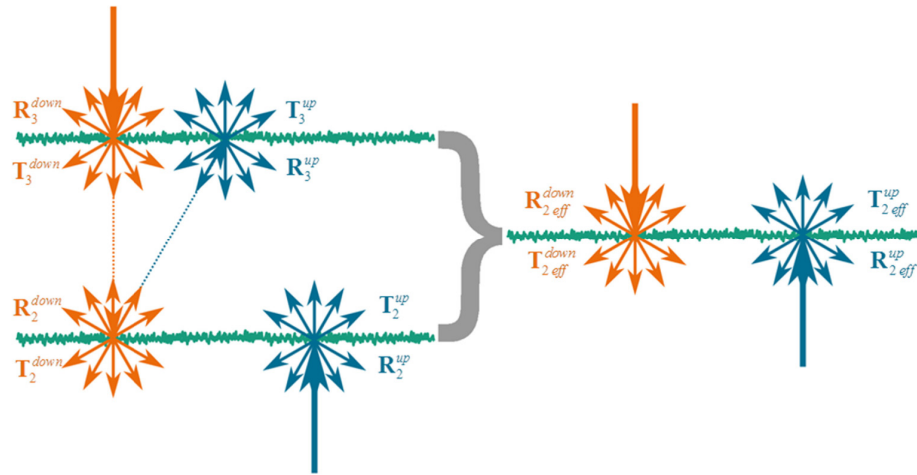


Fig. 2. Sketch of the effective interface (on the right), which exhibits the same redistribution properties as the layer with two textured interfaces (on the left). By combining interfaces in this manner, a system with multiple interfaces can be transferred to two or even one single effective interface.

The way to describe the physical mechanisms of light propagation and redistribution does not depend on the number of interfaces. The description for two interfaces was modified only with respect to the matrix name convention. It is most instructive to think of the total reflection as being composed of different parts. The first part is the direct reflection ($\mathbf{R}_3^{down}$),

the second has to be transmitted at the top interface ($\mathbf{T}_3^{down}$), propagate to ($\mathbf{D}_{glass}$) and be reflected at the middle interface ($\mathbf{R}_2^{down}$), propagate back ($\mathbf{D}_{glass}$) and be transmitted at the uppermost interface ($\mathbf{T}_3^{up}$). This is mathematically described by $\mathbf{T}_3^{up} \mathbf{D}_{glass} \mathbf{R}_2^{down} \mathbf{D}_{glass} \mathbf{T}_3^{down}$. The third and all further parts take additional cycles through the layer, $\mathbf{D}_{glass} \mathbf{R}_2^{down} \mathbf{D}_{glass} \mathbf{R}_3^{up}$, before the light is transmitted through the top surface. These considerations lead to the following expression for the effective interface reflection:

$$\mathbf{R}_{2\text{ eff}}^{down} = \mathbf{R}_3^{down} + \mathbf{T}_3^{up} \left[\sum_{i=0}^{\infty} (\mathbf{D}_{glass} \mathbf{R}_2^{down} \mathbf{D}_{glass} \mathbf{R}_3^{up})^i \right] \mathbf{D}_{glass} \mathbf{R}_2^{down} \mathbf{D}_{glass} \mathbf{T}_3^{down}, \quad (1)$$

where i indicates the number of cycles the light has to go back and forth inside the layer. Using the same approach for the transmission through the effective interface, again for light incident from above, gives the following expression:

$$\mathbf{T}_{2\text{ eff}}^{down} = \mathbf{T}_2^{down} \left[\sum_{i=0}^{\infty} (\mathbf{D}_{glass} \mathbf{R}_3^{up} \mathbf{D}_{glass} \mathbf{R}_2^{down})^i \right] \mathbf{D}_{glass} \mathbf{T}_3^{down}. \quad (2)$$

The relations for effective reflective and transmissive redistribution matrices of light incident from below can be calculated in the same manner. The only difference to Eq. (1) and Eq. (2) is the interface order and the incident direction. Hence, one can get the expressions by exchanging the interface numbers and the light direction of Eq. (1) and Eq. (2). This results in:

$$\mathbf{R}_{2\text{ eff}}^{up} = \mathbf{R}_2^{up} + \mathbf{T}_2^{down} \left[\sum_{i=0}^{\infty} (\mathbf{D}_{glass} \mathbf{R}_3^{up} \mathbf{D}_{glass} \mathbf{R}_2^{down})^i \right] \mathbf{D}_{glass} \mathbf{R}_3^{up} \mathbf{D}_{glass} \mathbf{T}_2^{up}, \quad (3)$$

$$\mathbf{T}_{2\text{ eff}}^{up} = \mathbf{T}_3^{up} \left[\sum_{i=0}^{\infty} (\mathbf{D}_{glass} \mathbf{R}_2^{down} \mathbf{D}_{glass} \mathbf{R}_3^{up})^i \right] \mathbf{D}_{glass} \mathbf{T}_2^{up}. \quad (4)$$

As described by the Eqs. above, the effective matrix calculation is just a multiplication and summation of matrices. The summation can be done either iteratively with a finite i_{max} or by using the Neumann series (geometric series for matrices). In the second case, the expressions could be rewritten in the following form:

$$\mathbf{R}_{2\text{ eff}}^{down} = \mathbf{R}_3^{down} + \mathbf{T}_3^{up} \left[\mathbf{I} - (\mathbf{D}_{glass} \mathbf{R}_2^{down} \mathbf{D}_{glass} \mathbf{R}_3^{up}) \right]^{-1} \mathbf{D}_{glass} \mathbf{R}_2^{down} \mathbf{D}_{glass} \mathbf{T}_3^{down}, \quad (5)$$

$$\mathbf{T}_{2\text{ eff}}^{down} = \mathbf{T}_2^{down} \left[\mathbf{I} - (\mathbf{D}_{glass} \mathbf{R}_3^{up} \mathbf{D}_{glass} \mathbf{R}_2^{down}) \right]^{-1} \mathbf{D}_{glass} \mathbf{T}_3^{down}, \quad (6)$$

$$\mathbf{R}_{2\text{ eff}}^{up} = \mathbf{R}_2^{up} + \mathbf{T}_2^{down} \left[\mathbf{I} - (\mathbf{D}_{glass} \mathbf{R}_3^{up} \mathbf{D}_{glass} \mathbf{R}_2^{down}) \right]^{-1} \mathbf{D}_{glass} \mathbf{R}_3^{up} \mathbf{D}_{glass} \mathbf{T}_2^{up}, \quad (7)$$

$$\mathbf{T}_{2\text{ eff}}^{up} = \mathbf{T}_3^{up} \left[\mathbf{I} - (\mathbf{D}_{glass} \mathbf{R}_2^{down} \mathbf{D}_{glass} \mathbf{R}_3^{up}) \right]^{-1} \mathbf{D}_{glass} \mathbf{T}_2^{up}, \quad (8)$$

where $\mathbf{I}$ is the identity matrix. The Neumann series leads to an analytically correct result with respect to the number of passes through the sheet. However, for very large matrices, the matrix inversion needs more computational resources than a large number of multiplications. Which of the methods is used can be chosen before the calculation.

2.3. System absorptance

As the three interfaces of Fig. 1 could be reduced to a sheet with only two interfaces, the absorptance of the bottom layer can be calculated as derived in [12,13] using the bottom surface as rear side and the effective interface as front side. The absorptance within the effective interface is considered automatically by lower entries for the transmittance. An alternative way to calculate the absorptance is to determine an absorptance matrix $\mathbf{Abs}_{bulk}$ for the layer under consideration as follows:

$$\mathbf{Abs}_{bulk} = \left(\sum_{i=0}^{\infty} (\mathbf{R}_{2\text{eff}}^{up} \mathbf{D}_{bulk} \mathbf{R}_1^{down} \mathbf{D}_{bulk})^i - \sum_{i=0}^{\infty} (\mathbf{D}_{bulk} \mathbf{R}_{2\text{eff}}^{up} \mathbf{D}_{bulk} \mathbf{R}_1^{down})^i \mathbf{D}_{bulk} + \sum_{i=0}^{\infty} (\mathbf{R}_1^{down} \mathbf{D}_{bulk} \mathbf{R}_{2\text{eff}}^{up} \mathbf{D}_{bulk})^i \mathbf{R}_1^{down} \mathbf{D}_{bulk} - \sum_{i=0}^{\infty} (\mathbf{D}_{bulk} \mathbf{R}_1^{down} \mathbf{D}_{bulk} \mathbf{R}_{2\text{eff}}^{up})^i \mathbf{D}_{bulk} \mathbf{R}_1^{down} \mathbf{D}_{bulk} \right) \mathbf{T}_{2\text{eff}}^{down} \quad (9)$$

The absorptance in the bulk layer is then given by

$$Abs_{bulk} = \sum_j \mathbf{Abs}_{bulk} \mathbf{v}_{inc}, \quad (10)$$

where j denotes the vector elements. In the same way as for the top layer, it is also possible to simplify the bottom layer to an effective interface. The effective matrix calculation, involving in this case interfaces 1 and 2, results in $\mathbf{R}_{1\text{eff}}^{up}$, $\mathbf{R}_{1\text{eff}}^{down}$, $\mathbf{T}_{1\text{eff}}^{up}$ and $\mathbf{T}_{1\text{eff}}^{down}$. The absorption matrix $\mathbf{Abs}_{glass}$ and the absorptance of the upper layer Abs_{glass} are calculated analogously to Eq. (9) and Eq. 10. The total module absorptance Abs_{total} is equal to the summation of the absorptance in the glass and bulk layer.

$$Abs_{total} = Abs_{bulk} + Abs_{glass} = \sum_j \mathbf{Abs}_{bulk} \mathbf{v}_{inc} + \sum_j \mathbf{Abs}_{glass} \mathbf{v}_{inc}. \quad (11)$$

2.4. System reflectance and transmittance

An elegant way to determine reflectance and transmittance of the total system is to take the effective upper surface and to further simplify it combined with the remaining bottom surface to one single effective interface, analogously to Eqs. (1)-(4). The reflectance R and transmittance T of the whole system can then be easily calculated using effective reflection and transmission matrices of this new interface $\mathbf{R}_{total}^{up}$, $\mathbf{R}_{total}^{down}$, $\mathbf{T}_{total}^{up}$ and $\mathbf{T}_{total}^{down}$, combined with the incoming light distribution:

$$R_{down} = \sum_j \mathbf{R}_{total}^{down} \mathbf{v}_{inc}, \quad (12)$$

$$T_{down} = \sum_j \mathbf{T}_{total}^{down} \mathbf{v}_{inc}. \quad (13)$$

The effective redistribution matrices for light incidence from below can be derived analogous to Eqs. (1)-(8). They can subsequently be used to calculate the corresponding absorptance, reflectance and transmittance. Altogether, OPTOS opens up the possibility of simulating monofacial or bifacial photovoltaic module stacks with an arbitrary number of textured material layers that can be coupled incoherently.

3. Exemplary effective redistribution matrix

The evaluation of redistribution matrices can reveal useful insights into the optical behavior of a textured interface, even without integrating it into a specific layer stack and calculating the complete system properties. The example matrices we present in Fig. 3 correspond to a silicon substrate [15] with random upright pyramid texture at a wavelength of 1100 nm, modeled using the in-house ray tracing tool Raytrace3D [16,17]. The random arrangement of the pyramids was simulated according to the geometrical model described in [2,18]. Random pyramids are the industrial standard front surface for monocrystalline silicon solar cells and therefore of high relevance. They exhibit a low broad band reflection, especially in combination with an anti-reflective coating (ARC), and furthermore show good light trapping properties [2,19].

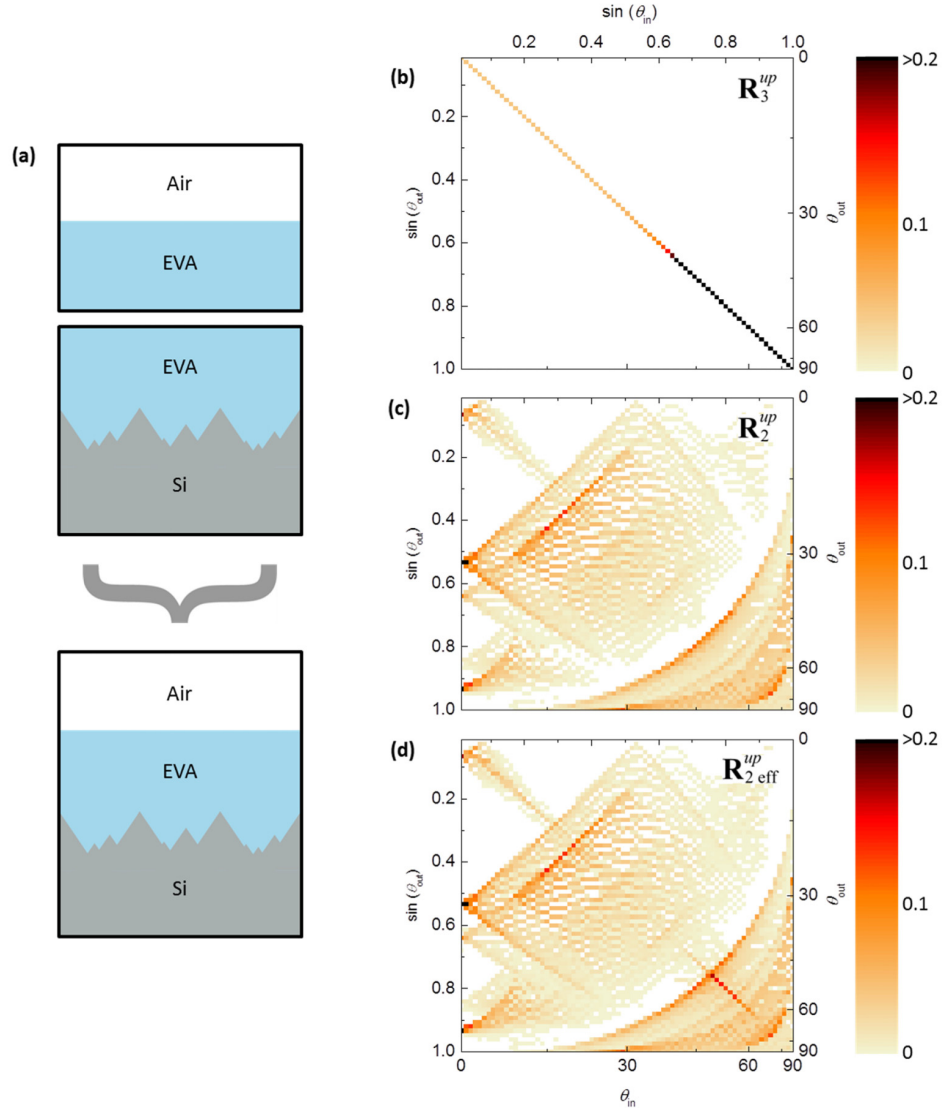


Fig. 3. Sketches of EVA-air, silicon-EVA and effective silicon-EVA-air interfaces in (a). The corresponding redistribution matrices, (b), (c) and (d) describe the internal reflection properties of light impinging onto the interface from below.

The module encapsulation consists of an EVA layer and a protective glass cover. Both of these layers are thick with respect to the considered wavelengths and can be treated incoherently. Glass and EVA exhibit very similar real parts of their refractive indices and very weak absorption for the wavelength range between ~ 400 nm and ~ 1150 nm. For simplicity, we neglect the absorption completely and treat EVA and glass as one single layer, without affecting the general optical behavior of the system. This encapsulation layer, which is called EVA from now on, has a planar interface toward the surrounding air. Figure 3(a) shows in the upper part the two-dimensional sketch of a planar EVA-air and a textured silicon-EVA interface with random pyramids and without ARC. Their internal reflectance properties are described by the redistribution matrices, (b) and (c), respectively. The combined interface is sketched in the lower part of (a) and represented by the effective matrix depicted in (d). Comparing matrices (c) and (d) shows larger values close to the diagonal for the effective system. This behaviour results from the specular internal reflection at the planar EVA-air interface. A large fraction of the light that is transmitted from the silicon to the EVA is reflected at the front surface and re-enters the silicon with the same angle.

The column sums of the matrices, which are depicted in Fig. 4, represent the probability for internal reflection at the corresponding interface for different polar angles. A larger value leads to long light path length within the solar cell. For each investigated interface, the internal reflection is almost unity for some angular regions and considerably reduced for other incident polar angles. Striking is the angular shift of the low reflectance region when the pyramids are encapsulated. Compared to the silicon-air interface (orange), the effective silicon-EVA-air interface (blue) shows the lowest reflection at polar angles between 30° and 50° instead of 40° to 70° . This behaviour is crucial when the front surface is to be combined with a rear surface that redistributes light preferably into certain polar angles. For example diffractive gratings could be designed to redirect incoming light into angles which have a high probability of total internal reflections at the front side. A rear surface that was optimized for its combination with a random pyramid front side on cell level and redirects most of the incoming light into angles just below 40° could induce a considerably lower absorption gain when the cell is encapsulated with EVA. Instead, a different optimization of the rear side structure for the module case has to be pursued.

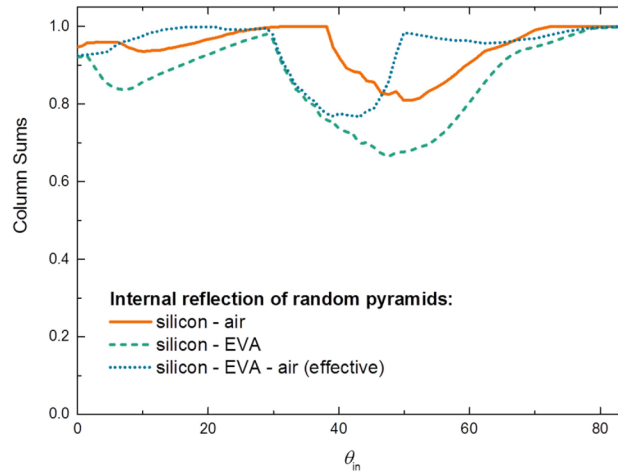


Fig. 4. Column sums of redistribution matrices (corresponding to internal reflectance) for interfaces with random pyramid texture. The polar angles with low reflectance exhibit a considerable angular shift when the surface is encapsulated in a higher refractive index material as EVA.

4. Optical module simulation

4.1. Normally incident light

As validation for the OPTOS module formalism, we investigate a system consisting of a 200 μm thick silicon sheet with random pyramid front side and perfect specular reflector at the rear side. Figure 5 shows the silicon absorptance and total reflectance of the described system, calculated with the OPTOS formalism. Results obtained with the PV-Lighthouse module ray tracing tool [20] are shown for comparison. Both simulations agree very well over the whole investigated wavelength range.

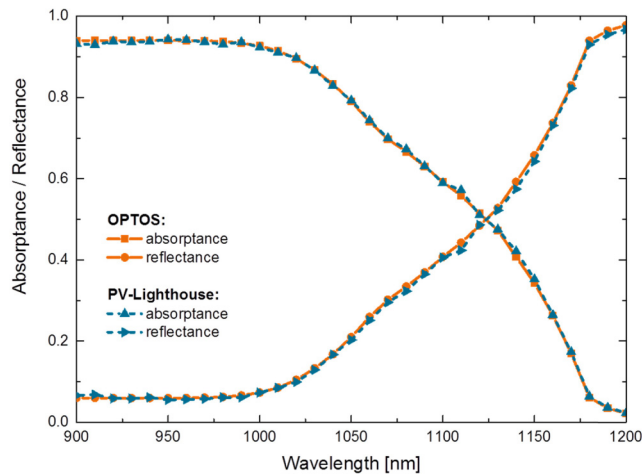


Fig. 5. Absorbance and reflectance of an encapsulated thick silicon sheet with random pyramids as front side texture and a perfect planar reflector at the rear side. The simulations results of the OPTOS module formalism and PV-Lighthouse ray tracing agree very well over the whole spectral range.

4.2. Diffractive grating integration: cell and module analysis

The modelling of systems with pyramidal front side and diffractive rear surface involves the efficient combination of ray and wave optical approaches, as demonstrated in [12,13]. Using the OPTOS module formalism enables the investigation of the optical cell to module loss for such systems. Figure 6 shows the absorptance of the front side textured silicon solar cell described above, but incorporating a 70 nm thick SiN_x anti-reflective coating [21], without encapsulation (dashed orange line) and with EVA (dashed blue line). Reflection at the planar air-EVA interface reduces the absorptance and causes an optical cell to module loss for most of the spectral range, whereas additional internal reflection at the same interface has the opposite effect (see Fig. 3).

The integration of a diffractive crossed grating at the rear side (silicon square pillars in contact with a planar perfect reflector of infinite refractive index, gaps filled with air, period: 1 μm , depth: 200 nm, fill factor: 0.5) causes a considerable absorptance gain for the cell and module configuration. The corresponding photocurrent increase is reduced from 0.8 mA/cm^2 to 0.6 mA/cm^2 by the EVA encapsulation for this specific grating. The reduction is partly due to the generally lower front surface transmittance, which weakens the effect of an additional rear side texture, and partly due to the improved light trapping behavior of the module without grating for wavelengths above ~ 1060 nm.

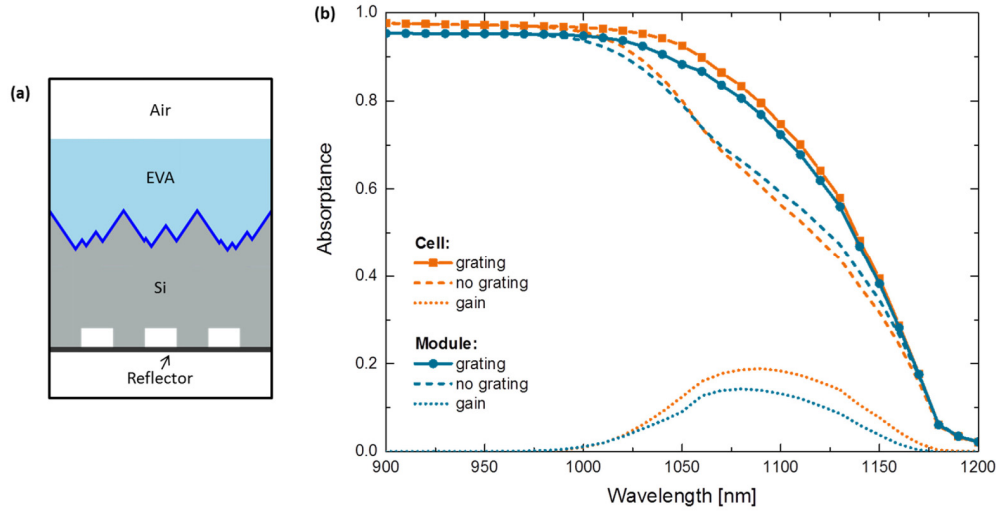


Fig. 6. (a) Sketch of the encapsulated solar cell with pyramidal front side texture including an SiNx ARC and a diffractive backside grating (structure dimensions are not to scale). (b) Absorbance investigation for systems with and without diffractive grating at the rear side and random pyramids at the front side. In both cases, with (blue) and without the encapsulation (orange), the grating enhances the solar cell absorbance.

4.3. Incidence angle variation

Variations of the angle of incidence play a major role for the yield analysis of silicon based photovoltaic modules. Using OPTOS, the module absorbance for different angles of incidence can be determined within minutes on a standard desktop PC. The same efficient matrix multiplication procedure can be applied for all angles. Only the incident power vector $\mathbf{v}_{inc}$ needs to be changed. Figure 7 shows the silicon absorbance for the encapsulated system described above, with and without diffractive grating at the rear.

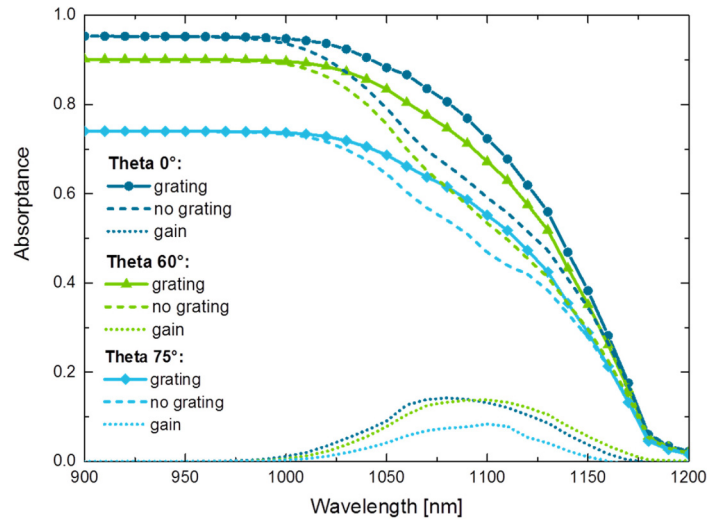


Fig. 7. Incidence angle variation for encapsulated silicon solar cells with and without diffractive grating at the rear side and random pyramids at the front side. For incidence angles up to 60°, the grating induced absorbance gain is almost not affected.

Increasing the polar angle of the incident light results in a higher reflection at the planar air-EVA interface. This is the dominant effect determining the lower silicon absorptance. It is comparably small for angles between 0° and 60° and increases for larger incident angles. The absorptance gain induced by the diffractive grating is almost not affected for incidence angles up to 60° .

5. Summary

In this paper, we introduced the matrix-based OPTOS formalism for systems consisting of an arbitrary number of plane-parallel, optically active interfaces. The mathematical expressions for the combination of two textured surfaces enclosing a thick, homogeneous material layer to one effective interface were derived. Subsequently, OPTOS was applied to the optical modelling of photovoltaic module stacks incorporating textured silicon cells and an EVA encapsulation. Already the evaluation of the reflection redistribution matrices allows useful insights into the optical properties of an interface, as shown for the random pyramid texture. Compared to the silicon-air interface, the effective silicon-EVA-air interface shows the lowest reflection at polar angles between 30° and 50° instead of 40° to 70° . This paves way for design rules of diffractive rear side textures with regard to their application in photovoltaic modules. The comparison of silicon absorptance and module reflectance of a comparably simple system, using OPTOS and an alternative simulation technique based on pure ray tracing, showed a very good agreement. The integration of a specific diffractive rear side grating into encapsulated solar cells with pyramidal front side texture showed a slightly lower gain compared to a cells without encapsulation. The variation of the angle of incidence had almost no effect on the grating gain for polar angles up to 60° . Both investigations demonstrate the versatility and efficiency of the formalism, which makes OPTOS a very suitable tool for the optical analysis of photovoltaic modules.

Acknowledgments

The research leading to these results has received funding from the German Federal Ministry for the Environment, Nature Conservation and Nuclear Safety under contract number 0325292 “ForTeS”. N. Tucher gratefully acknowledges the scholarship support from the Cusanuswerk, Bischöfliche Studienförderung. J. Eisenlohr gratefully acknowledges the scholarship support from the Deutsche Bundesstiftung Umwelt DBU.